Answers and talk notes

Volume of Rectangular Prisms

What your student is learning

Tonight we compute the volume of several boxes — fractional edges and all — and use the results to choose one.

Answers

Ask your student to explain before you look.

1. **B.** 24 m^3. 24 m^3: 3 × 2 × 4 = 24.
2. **A.** 15 ft^3. 15 ft^3: 2 × 5 = 10, and 10 × 1.5 = 15.
3. **C.** Box P, 9 cubic units to 8 cubic units. Box P, 9 cubic units to 8 cubic units: P is 3 × 2 × 1.5 = 9 and Q is 2 × 2 × 2 = 8.
4. **D.** 5 × 2 × 2.5. 5 × 2 × 2.5 is 25 cubic feet, which is more than 20.
5. **B.** Box M, by 7 cubic units. Box M, by 7 cubic units: M is 6 × 3 × 1.5 = 27, N is 5 × 4 × 1 = 20, and 27 − 20 = 7.

Talk about it

Which box holds more, and how much more does it hold?

Home mission

Pick two boxes, like a cereal box and a snack box. Measure each edge to the nearest half inch. Find each volume without rounding. Which holds more?

When they are stuck, ask

- Did you use the exact edge, or did you round it?
- How many edges go into a volume?
- What number does the problem say we have to reach?

Try not to say

- Round the decimal first, it is easier that way.
- Pick the biggest box, that is always the answer.
- Square units are fine for a box.